Raspberry Pi Laparoscope

Mechatronic Project 478
Proposal

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[1] Position Statement on Ethical use of Artificial Intelligence in Research and Teaching-Learning Assessment

[2] Draft interim SU guidelines on allowable AI use and academic integrity in assessment

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Chapter 1

Literature Review

A literature review has been conducted for this project. This section of the report documents the relevant research performed and divides into to the following sections:

- **Laparoscopy Overview:** An overview of what laparoscopy is and its history.
- **Current Technological Advancements:** The current directions that modern laparoscopy is improving on.
- **Current Available Technology:** A summary of current Laparoscopes that is available in South Africa.
- **Available Hardware Components:** An overview of different hardware modules available for this project.
- **Software Considerations:** An brief overview of the available programs that can be used for this project.
- **Visual Inspection Considerations:** An overview of visual inspection and aspects of camera that need to be mastered.

1.1 Laparoscopy Overview

Laparoscopy is a branch of the endoscopic medical field. It has become an integral part of surgical procedures in all field in the recent decades. Inspired by the minimally invasive surgery (MIS) approach, laparoscopic surgery consist of making 1 or more small incisions on the abdomen of the patient. Surgeons use these incision to insert thin rod medical equipment into the patient to perform the surgery. (Ibrahim *et al.*, 2021)

An essential tool of this surgical procedure is the laparoscope. Due to small size of the incisions, the surgeons' view is limited. The laparoscope is the solution to this problem. It allows the surgeons to view what is occurring within

the patient while the operation is being performed. Therefore, they can clearly manipulate their equipment from outside the patient to perform complex procedures inside the patient. (Subido *et al.*, 2018)

There are 2 design methods being used currently when developing laparoscopes. Their main difference is the placement of the digital camera on the tool itself. The first method is inspired by the *original laparoscope design*. This method utilized reflective lenses and cold light sources, which allowed the surgeon to inspect the abdomen of a patient, similar fashion as a microscope. However, instead of an eye piece, a camera is used so that the captured digital images can be displayed in real-time to the surgical staff.(Ibrahim *et al.*, 2021)

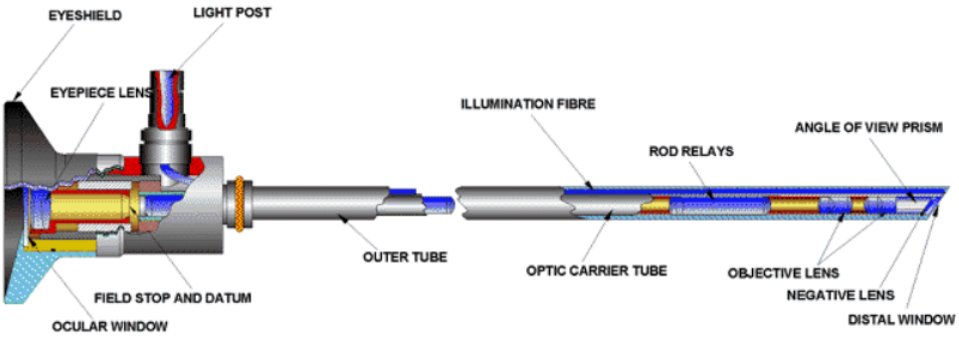

Figure 1.1: Original laparoscope design. (ebme.co.uk, 2026)

The second design method use to develop laparoscope utilizes chip on stick technology. (Mishra, 2026) This design method places the camera of the laparoscope on the tip of the tool. This method removes the complex lenses required for the traditional and use a variety of wide lens camera modules. (Subido *et al.*, 2018)
Laparoscopic surgery has been one of the greatest advancements in medical industry in recent years. *Buia* et al. *(2015)* shows that laparoscopic surgery has quicker recovery time and less pain after surgery compared to traditional open surgery. However, it shows that certain procedures are more time-consuming and more complex to perform.

1.2 Current Technological Advancements

Since the addition of cameras to laparoscopes, there has been a large addition of advancement available. These advancement range from hardware improvements to additional software programming abilities to the existing design.

The main hardware advancement made to the camera module is increasing the field of view (FoV). However, there are additional advancements made in hardware for laparoscopes. The addition of robotics to the laparoscopes has greatly

assisted the laparoscopic surgical industry. Robotic manipulators are utilized to assist surgeons by maintaining stable control of the laparoscope, therefore, ensuring consistent and clear visualization during the procedure.(Subido *et al.*, 2018)

There are multiple additions currently being made to the software of laparoscopes. *Subido* et al. *(2018)* list the following additions:

- Digital imaging enhancement techniques.
- AI incorporation.
- 3D visualization.
- Object tracking.

1.3 Current Available Technology

There is a range of different laparoscope or endoscope suppliers. Each of these suppliers produce more than 1 product. Therefore, only suppliers for South African hospitals will be reviewed and Intuitive Surgical that produces the *Da Vinci* system, that has been dominating the surgical robot market. (Subido *et al.*, 2018)

One of the main suppliers of laparoscopic equipment in South Africa is Karl Storz Endoskope, a German company, with a sales office in Cape Town. They produce various products in the medical industry. (karlstorz.com, 2026*a*) One of their products is the *TIPCAM1 RUBINA*.

TIPCAM1 RUBINA is a rigid endoscope with the ability of 0° and 30° viewing angle. It also comes with automatic horizon control and the ability to easily swap between 2D and 3D imaging. Its has an outer diameter of 10.3 mm and a working length of 320 mm. It is has 4K image resolution and has a mass of 420 g. (karlstorz.com, 2026*b*)

Figure 1.2: Figure of the TIPCAM1 RUBINA. (karlstorz.com, 2026*b*)

Olympus Corporation is another company that supplies medical equipment globally. They are a Japanese manufacturing company that produces the *ENDOEYE 3D 10 mm*. The *ENDOEYE 3D 10 mm* is a video telescope, with the ability of 3D imaging at a viewing angle of 0° and 30°. It is able to perform image rotation at 30° viewing angle, while maintaining the horizon. (olympus.com.au, 2026)

Figure 1.3: Figure of the ENDOEYE 3D 10 mm. (olympus.com.au, 2026)

Certain international companies uses local distributors to sell their products. Intuitive Surgery, an American company, uses this method. They designed and manufacture the *Da Vinci* medical robot. This system has various variants for different surgical purposes. Currently, there is *Da Vinci Xi* and *Da Vinci X* for multi port surgery. However, the *Da Vinci SP* is designed for single port surgery. These robot systems are are equipped with 7 DOF movement and 3D visualization. Their only drawback is there large size and enormous financial procurement cost. (intuitive.com, 2026)Below is a figure of the *Da Vinci SP*.

Figure 1.4: Figure of the Da Vinci SP. (intuitive.com, 2026)

1.4 Available Hardware Components

There multiple hardware components required to build a laparoscope. Therefore, this section will be divided into multiple subsection where different products will be reviewed for each component. These subsections include:

- Single board computer module.
- Camera module.

1.4.1 Single Board Computer Module

Single board computers (SBC) is a credit-card-size computer, able to perform complex computations while operating various systems. For the laparoscope, it will be the main "brain" of the system. Therefore, it is critical that it can perform all necessary tasks it requires. There is 2 main suppliers of SBC:

- Raspberry Pi Ltd.
- Arduino AG.

Raspberry Pi has a large range of products that is available for the project. However, Raspberry Pi's SBCs has an important feature installed into most of SBC, that would be beneficial to the project. This feature is the designated camera serial interface (CSI) or the MIPI camera display transceivers, according to Raspberry Pi. This feature allows for high-quality videos at high speeds. (raspberrypi.com, 2026*a*) The *table* below show the available options of all the Raspberry Pi SBC:

Table 1.1: Raspberry Pi SBCs.(raspberrypi.com, 2026*b*)

Raspberry Pi SBC	RAM	CPU Speed	MIPI Ports	Cost
Pi 5	1-16 GB	2.4 GHz	2	R 3 266.00
Pi 4 Model B	1-8 GB	1.5 GHz	1	R 1 360.00
Pi 3 Model B+	1 GB	1.4 GHz	1	R 640.43
Pi Zero 2 W	512 MB	1 GHz	2	R 240.16
Pi Zero 2	512 MB	1 GHz	2	R 309.90

All of the SBC in *table* above have these features:

- 40 GPIO Pins.
- Wi-Fi Capabilities.
- Bluetooth Capabilities.

Arduino, similar to Raspberry Pi, have a wide range of products available for this project. However, they have one absent feature, namely the lack of CSI feature. Therefore, they are incompatible with the Raspberry Pi Camera Modules. However, they can use other camera modules, with the use of SPI and I2C GPIO pins. Although this options isn't favourable, due to trade-off in quality during longer videos, it is still a feasible option that can be explored. (Public, 2022) The *table* below shows the available products from Arduino:

Table 1.2: Arduino SBCs. (store.arduino.cc, 2026)

Arduino SBC	RAM	CPU Speed	SPI Ports	Cost
UNO R4 WiFi	32 kB	48 MHz	1	R 488.46
UNO R4 Minima	32 kB	48 MHz	1	R 352.33
UNO R3	2 kB	16 MHz	1	R 469.24
GIGA R1 WiFi	1 MB	480 MHz	2	R 1 247.58
Nano ESP32	512 kB	240 MHz	1	R 358.74
Nano RP2040 Connect	264 kB	133 MHz	4	R 389.17
UNO Q	2 GB	2 GHz	1	R 1 035.54
Nano 33 BLE	256 kB	64 MHz	4	R 456.43
Nano 33 IoT	520 kB	240 MHz	6	R 427.6
Nano Every	6 kB	20 MHz	1	R 232.22

1.4.2 Camera Module

The camera module is another important component of this project. It is essential to capture the real-time images inside the patient. Therefore, this component should be selected carefully. A common occurrence with the camera modules is that the size of the camera sensor and assembly can fit in a Ø10 mm hole. However, the circuit board that it is attached to it does not fit inside this hole. Therefore, options with removable camera assemblies that can be attached to the circuit board with a wire connection were reviewed. Consequently, camera modules that already fit inside with the hole, that comes with a wire extension is reviewed as well.

Raspberry Pi camera modules all have all circuit board connection, that can't separate from the camera sensor, as can be seen in the *figure* below. However, it is optional to purchase the stand-alone sensor assembly for the Raspberry Pi camera module V3. Although, this camera sensors it powerful, with IMX708 sensor capable of 4608 x 2592 pixels images, it physical size is 10.8 mm x 10.8 mm. Therefore, it is incapable to fit inside the Ø10 mm hole. (arducam.com, 2020)

As mention in the previous section, *Single board computer module*, the Rasp-

Figure 1.5: Figure of Raspberry Pi Camera Module 3

berry Pi SBC modules have CSI ports. Even though Raspberry Pi camera modules can't be considered for this project, other suppliers make camera modules that can interface with these CSI ports. Below is a *table* of the supplier, Leopard Imaging, from their hummingbird mobile camera modules series:

Table 1.3: Leopard Imaging camera modules.(leopardimaging.com, 2026)

Product code	Resolution	Max fps	FoV (°)	Focus depth
LI-OV2685-MIPI-FF	2 MP	120	60	30 cm - inf
LI-OV8865-MIPI-AF	8 MP	60	70	10 cm - inf
LI-OV13850-MIPI-AF	13 MP	30	70	10 cm - inf
LI-OV5640-MIPI-AF	5 MP	120	65	10 cm - inf
LI-AR1335C-MIPI-061H-AF	13 MP	30	61	10 cm - inf
LI-IMX258-MIPI-AF	13 MP	60	79.8	10 cm - inf
LI-IMX586-MIPI-AF-79D	48 MP	240	79	N/A
LI-IMX681-MIPI-AF-96D	12 MP	30	96	N/A
LI-AR0246-FLEX-80H	2 MP	60	80	35 cm - inf
LI-OV16E10-AF-D78	16 MP	120	78.4	7 cm - inf
LI-AR0544-MIPI-150H	5 MP	60	150	15 cm - inf
LI-OS04C10-MIPI-107H	4 MP	240	107	70 cm - inf
LI-OG12B10-MIPI-AF-95D	12 MP	240	75	10 cm - inf
LI-OV13B10-MIPI-AF-75D	13 MP	60	75	10 cm - inf

Another camera module manufacturer is ArduCam Technology Co., Ltd. They specialize in embedded camera modules and is a leading company in the designing of camera modules. They also produce several camera modules, however,

their most relevant product to this project is their *OV5647 Spy Camera module*. Here is a list of the products specifications:

- Resolution: 5 MP.
- Max fps: 90.
- FoV: 120°.
- Cost: R 316.89

This product however does have one flaw. Its focus depth is 1 m to inf.(store.arduino.cc, 2026) Therefore, this product cannot be used for this project.

Figure 1.6: Figure of Arducam spy camera module.(store.arduino.cc, 2026)

For this project, there is an option of purchasing endoscope camera modules. These camera modules are designed to be used in a endoscopic application. There are various designers and manufacturers and each of these have their own series of products.

IADIY Technology is a Chinese company, that creates a line of products that can communicate with USB cable attached to the camera module. Below is a *figure* of a IADIY camera module. These camera modules are powerful, with their compact size, wide Field-of-View angles and built in light sources. Here is a *table* that list all the available products from IADIY technology

Figure 1.7: Figure of IADIY camera module.(iadiy.com, 2026)

Table 1.4: IADIY endoscopic camera modules. (iadiy.com, 2026)

Product Code	Resolution	Ø of Module (mm)	FoV (°)	Max fps
SCM03M30D5.5U	0.3 MP	5.5	60	30
SCM1M30D5.5U	1 MP	5.5	88	30
SCM1M30D3.9U	1 MP	3.9	140	30
SCM2M30D8U	2 MP	8	80	30

Guangzhou Sincere Information Technology Ltd. (SincereFirst is their trade name) is another Chinese company. They specialize in various camera products that can be used through USB connection. (cameramodule.com, 2026) Below is a *table* of all the available products from SincereFirst.

Table 1.5: SincereFirst endoscopic camera modules(sincerefirst.com, 2026)

Image sensor	Resolution	Ø of Module (mm)	FoV (°)	Max fps
OmniVision OCHFA20	720 P	1.05	120	30
OmniVision OCHFA10	0.5 MP	1.5	105	30
N/A	0.5 MP	2	120	30
Brigates ES101	1 MP	3.1	140	75

Motoshot Technology Co., Ltd. (Camemake is their trading name) is a Chinese company. Since their establishment in 2016, they have been designing and manufacturing camera modules that can interface with USB, DVP/SPI and CSI. They have 3 endoscopic products that can be considered for this project, as well as a range of other camera module.(camemake.eu, 2026*b*) Below is a *table* that shows some of the products specifications:

Table 1.6: Camemake endoscopic camera modules. (camemake.eu, 2026*a*)

Sensor	Resolution	Ø of Module (mm)	FoV (°)	Max fps	Focus Depth
N/A	1080 P	7	60	30	30 - 50 mm
OV2640	2 MP	1.6	120	30	5 - 50 mm
OV9734	720 P	4.2	100	30	5 -100 mm

All of these products mentioned above in endoscopic camera module tables, all have built in illumination modules. Thus, minimizing the amount of work required to design and manufacture the developed laparoscope.

1.5 Software Considerations

This project is mainly a design problem that has to be addressed. There are however software application needed for this project. The most important ap-

plication is that of the camera module interface.

The software should enable the SBC to connect and operate the camera module. raspberrypi.com (2026*a*) properly documents how to use a camera module that interfaces with an CSI connection. If a SPI/I2C connection is required to interface with the camera module, Public (2022) is library that describes how it can be performed. If a USB connection is used as an interface between the SBC and camera module, then minimal action is required to interface the 2 modules. (mechatronicslab.net, 2026)

There are however other applications required for this project. As mentioned in the software component of the scope, there are 2 other section of code that need to be considered for this project. The ability display the captured footage to a real-time monitor, depends on the monitor that will be used for this project. If a personal computer is used, the captured footage can be transferred with the means of WiFi, Bluetooth, USB connection, or even a Mathworks ThingSpeak channel.(uk.mathworks.com, 2021)

Another important aspect to considered is coding language used for this project. However, this is dependent of the SBC used. If a Raspberry Pi is used, then Python is recommended language to use. (raspberrypi.com, 2026*a*) However, if Arduino is used then the ideal language to use is either C or C++. (docs.arduino.cc, 2026) Visual Studio Code (VS Code) can be used to perform all programming for this project.

Documentation of this project will be performed using LaTeX. VS Code will be used to run this application, through the LaTeX Workshop extension.(Yu, 2026)

1.6 Visual Inspection Considerations

This project requires a visual camera to be a functional laparoscope. However, there is various factors to be considered to take high-quality images. This section of the literature reviews aims to explain each of these factors. Consequently, how they can impact this project.

Important concept to understand, is how a digital image is captured. A camera sensors consists of a matrix of millions of light cavities. When a picture is being taken, these cavities, called ”photosites”, are exposed to the photons of the object or scene. The photons are then collected by the photosites and stored as an electrical charge. Consequently, when the image is finished being taken, these charges are quantified and then processed to make the image. (cambridgeincolour.com, 2020*b*)

One aspect that needs to be mastered to captured quality images is the exposure

triangle. It is 3 pillars that dictates the quality of an image. Aperture determines the size of the area opened in the lens which light can filter through to sensor. The larger the aperture value, measured in f-stop values, the smaller the hole size where light filters through the lens. This pillars relates to the depth of field of the images. The smaller aperture correlates to shallower depth of field images. (cambridgeincolour.com, 2020*a*)

Shutter speed is the second pillar to the exposure triangle. When a digital images is being taken the lens opens up to expose the photosites to the light. Consequently, shutter speed refers to total duration of the lens being open, thus causing the sensor to be exposed to image. If the shutter speed increases, then the sensors is exposed to reduced light from the images. Therefore, as the shutter speed increases, the images will gain more quality, with more definitive objects that is more "frozen" in the image. (cambridgeincolour.com, 2020*a*)

The third pillar is ISO. This determines how sensitive the sensor is to the exposed light. As the ISO value increases, the higher the sensitivity value will become. Therefore, the image will have increased levels of noise captured if the ISO value is increased. (cambridgeincolour.com, 2020*a*)

Focal length is a critical feature that has to be considered for this project. Focal length, is the distance between the camera sensor and the camera lens. As this value increases, then the field of view value will decrease. Field of view, measured in degrees, is essentially how much of an images can be seen. If this value increases, then a wider image can be taken. (Morrow, 2026)

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